



Practical File Of Problem Solving using Python Programming 24CSE0101

Submitted

in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE & ENGINEERING



CHITKARA UNIVERSITY

**CHANDIGARH-PATIALA NATIONAL HIGHWAY
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December, 2024

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1st Sem, 2024

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Program 1: a) Write a Python Program to Calculate the Area of a Triangle

```
# Program to calculate the area of a triangle
def area_of_triangle(base, height):
    return (base * height) / 2

# Example usage
base = 10
height = 5
result = area_of_triangle(base, height)
print("Area of the triangle:", result)
```

Program 1. b) Write a Python Program to Swap Two Variables

```
# Program to swap two variables
def swap_variables(a, b):
    return b, a

# Example usage
x = 5
y = 10
x, y = swap_variables(x, y)
print("Swapped variables:", x, y)
```

Output 1: a)

```
Area of the triangle: 25.0
```

Output 1: b)

```
Swapped variables: 10 5
```

Program 1. c) Write a Python Program to Convert Celsius to Fahrenheit

```
# Program to convert Celsius to Fahrenheit
def celsius_to_fahrenheit(celsius):
    return (celsius * 9/5) + 32

# Example usage
temp_celsius = 25
temp_fahrenheit = celsius_to_fahrenheit(temp_celsius)
print("Temperature in Fahrenheit:", temp_fahrenheit)
```

Output 1: c)

```
Temperature in Fahrenheit: 77.0
```


Program 2: a) Write a Python Program to check if a number is odd or even

```
def check_odd_even(number):  
    return "Even" if number % 2 == 0 else "Odd"  
# Example usage  
num = 7  
result = check_odd_even(num)  
print("The number is:", result)  # The number is Odd or Even
```

Program 2. b) Write a Python Program to Check if a Number is Positive, Negative or 0

```
def check_number(num):  
    if num > 0:  
        return "Positive"  
    elif num < 0:  
        return "Negative"  
    else:  
        return "Zero"  
  
# Example usage  
number = -5  
result = check_number(number)  
print("The number is:", result)
```

Output 2: a)

```
python -u "e:\Chitkara University\Uni  
The number is: Odd
```

Output 2: b)

```
The number is: Negative
```

Program 2: c) Program to check if a number is an Armstrong number

```
def is_armstrong(num):  
    order = len(str(num))  
    sum_of_powers = sum(int(digit) ** order for digit in str(num))  
    return num == sum_of_powers  
  
# Example usage  
number = 153  
result = is_armstrong(number)  
print("Is Armstrong number?", result)
```

Output 2: c)

```
Is Armstrong number? True  
PS E:\Chitkara University>
```

Program 3. a) Write a Python program to check if a given number is a Fibonacci number.

```
# Program to check if a number is Fibonacci
```

```
def is_fibonacci(n):
```

```
    a, b = 0, 1
```

```
    while a < n:
```

```
        a, b = b, a + b
```

```
    return a == n
```

```
# Example usage
```

```
num = 5
```

```
result = is_fibonacci(num)
```

```
print("Is Fibonacci number?", result)
```

Program 3: b) Write a Python program to print the cube sum of the first n natural numbers.

```
# Program to calculate the cube sum of first n natural numbers
```

```
def cube_sum(n):
```

```
    return sum(i**3 for i in range(1, n + 1))
```

```
# Example usage
```

```
n = 3
```

```
result = cube_sum(n)
```

```
print("Cube sum of first", n, "natural numbers:", result)
```



Output 3: a)

```
iversity\Uni Python\new.py"  
Is Fibonacci number? True
```

Output 3: b)

```
Cube sum of first 3 natural numbers: 36
```

Program 3: c) Write a Python program to print all odd numbers in a range.

```
# Program to print all odd numbers in a range

def print_odd_numbers(start, end):

    return [num for num in range(start, end + 1) if num % 2 != 0]

# Example usage

start = 1

end = 10

result = print_odd_numbers(start, end)

print("Odd numbers in the range:", result)
```

Output 3: c)

```
Odd numbers in the range: [1, 3, 5, 7, 9]  
PS E:\Chitkara University>  
Spaces: 4 UTF-8 CRLF {} Python 3.12.5 64-bit (system) Abi
```


Problem 4. a) Write a Python Program to Print Pascal's Triangle

```
# Program to print Pascal's Triangle
def pascal_triangle(n):
    triangle = []
    for i in range(n):
        row = [1] * (i + 1)
        for j in range(1, i):
            row[j] = triangle[i - 1][j - 1] + triangle[i - 1][j]
        triangle.append(row)
    return triangle

# Example usage
n = int(input("Enter number of rows:"))
result = pascal_triangle(n)
for row in result:
    print(row)
```

Program 4. b) Write a Python Program to Draw the Following Pattern

```
# Program to draw a specific pattern
def draw_pattern(n):
    for i in range(1, n + 1):
        print((str(i) + ' ') * (n - i + 1))

# Example usage
n = int(input('Enter n: '))
draw_pattern(n)
```



Output 4: a)

```
> python d...
iversity\Uni Python\new.py"
Enter number of rows:4
[1]
[1, 1]
[1, 2, 1]
[1, 3, 3, 1]
```

Output 4: b)

```
> python d...
iversity\Uni Python\new.py"
Enter n: 5
1 1 1 1 1
2 2 2 2
3 3 3
4 4
5
PS E:\Chitkara University>
Spaces: 4 LTF: 8 CRLF: {} Python: 3.12.5 64-bit (over
```

Program 5. Write a program with a function that accepts a string from keyboard and creates a new string after capitalizing each word.

```
# Program to capitalize each word in a string
def capitalize_words(sentence):
    return ' '.join(word.capitalize() for word in sentence.split())
# Example usage
input_string = input("Enter the string: ")
result = capitalize_words(input_string)
print("Capitalized string:", result)
```

Program 6. a) Write a program that accepts a list from the user, reverses the content of the list, and displays it. Do not use the reverse() method.

```
# Program to reverse a list without using reverse()
def reverse_list(lst):
    return lst[::-1]

# Example usage
user_input = input("Enter numbers separated by spaces: ")
input_list = list(map(int, user_input.split()))
reversed_list = reverse_list(input_list)
print("Reversed list:", reversed_list)
```

Output 5:

```
Enter the string: stop and smell the roses  
Capitalized string: Stop And Smell The Roses
```

Output 6: a)

```
Enter numbers separated by spaces: 1 2 3 4 5  
Reversed list: [5, 4, 3, 2, 1]
```

Program 6. b) Find and display the largest number of a list without using the built-in function max().

```
# Program to find the largest number in a list without using max()
def find_largest(lst):
    largest = lst[0]
    for num in lst:
        if num > largest:
            largest = num
    return largest

# Example usage
user_input = input("Enter numbers separated by spaces: ")
input_list = list(map(int, user_input.split()))
largest_number = find_largest(input_list)
print("Largest number:", largest_number)
```

Problem 7. Find the sum of each row of a matrix of size m x n.

```
# Program to find the sum of each row of a matrix

def sum_of_rows(matrix):
    return [sum(row) for row in matrix]

# Example usage
matrix = [
    [1, 2, 3],
    [4, 5, 6],
    [7, 8, 9]
]
row_sums = sum_of_rows(matrix)
print("Sum of each row:", row_sums)
```

Output 6: b)

```
Enter numbers separated by spaces: 3 1 4 1 5  
Largest number: 5  
PS E:\Chitkara University>
```

Output 7:

```
iversity\Uni Python\new.py"  
Sum of each row: [6, 15, 24]
```

Problem 8. a) Write a program that reads a string from keyboard and displays the number of uppercase letters, lowercase letters, digits, and whitespace characters.

```
# Program to count different character types in a string
def count_character_types(string):
    uppercase = sum(1 for c in string if c.isupper())
    lowercase = sum(1 for c in string if c.islower())
    digits = sum(1 for c in string if c.isdigit())
    whitespace = sum(1 for c in string if c.isspace())
    return uppercase, lowercase, digits, whitespace

# Example usage
input_string = input("Enter a string: ")
result = count_character_types(input_string)
print("Uppercase letters:", result[0])
print("Lowercase letters:", result[1])
print("Digits:", result[2])
print("Whitespace characters:", result[3])
```

Problem 8. b) Write a Python Program to Find Common Characters in Two Strings.

```
# Program to find common characters in two strings
def common_characters(str1, str2):
    return set(str1) & set(str2)

# Example usage
string1 = "hello"
string2 = "world"
result = common_characters(string1, string2)
print("Common characters:", result)
```

Output 8: a)

```
Enter a string: Hello World 123!  
Uppercase letters: 2  
Lowercase letters: 8  
Digits: 3  
Whitespace characters: 2
```

Output 8: b)

```
Common characters: {'o', 'l'}
```


Problem 8. c) Write a Python Program to Count the Number of Vowels in a String.

Program to count the number of vowels in a string

```
def count_vowels(string):  
    vowels = "aeiouAEIOU"  
    return sum(1 for char in string if char in vowels)
```

Example usage

```
input_string = "Hello World"  
result = count_vowels(input_string)  
print("Number of vowels:", result)
```

Output 8: c)

```
Number of vowels: 3  
PS E:\Chitkara University>
```

Problem 9: a) Write a Python program to check if a specified element is present in a tuple of tuples.

```
# Program to check if an element is in a tuple of tuples
def check_in_tuples(tuples, element):
    for tup in tuples:
        if element in tup:
            return True
    return False

# Example usage
tuple_of_tuples = (('Red', 'White', 'Blue'), ('Green', 'Pink', 'Purple'), ('Orange', 'Yellow', 'Lime'))
result = check_in_tuples(tuple_of_tuples, "White")
print("Is the element present?", result)
```

Problem 9. b) Write a Python program to remove empty tuple(s) from a list of tuples.

```
# Program to remove empty tuples from a list
def remove_empty_tuples(tuple_list):
    #In Python, empty tuples () are considered False in a boolean context, while non
    #empty tuples are True.
    return [tup for tup in tuple_list if tup]

# Example usage
list_of_tuples = [(), (), ('), ('a', 'b'), ('a', 'b', 'c'), ('d')]
result = remove_empty_tuples(list_of_tuples)
print("List after removing empty tuples:", result)
```

Output 9: a)

```
PS E:\Chitkara University> python -u "e:\Chitkara University\Uni Python\new  
Is the element present? True
```

Output 9: b)

```
List after removing empty tuples: [(' '), ('a', 'b'), ('a', 'b', 'c'), 'd']  
PS E:\Chitkara University> 
```

Problem 10. a) Write a Program in Python to Find the Differences Between Two Lists Using Sets.

```
# Program to find differences between two lists using sets
def list_differences(list1, list2):
    return list(set(list1) - set(list2)), list(set(list2) - set(list1))
# Example usage
list1 = [1, 2, 3, 4]
list2 = [3, 4, 5, 6]
diff1, diff2 = list_differences(list1, list2)
print("Difference in list1:", diff1)
print("Difference in list2:", diff2)
```

Problem 11. a) Write a Python program to remove duplicate values across Dictionary Values.

```
# Program to remove duplicate values across dictionary values
def remove_duplicates(dict_values):
    seen = set()
    result = {}
    for key, values in dict_values.items():
        result[key] = []
        for value in values:
            if value not in seen:
                seen.add(value)
                result[key].append(value)
    return result

# Example usage
sample_dict = {'Manjeet': [1, 1, 1], 'Akash': [1, 1, 1]}
result = remove_duplicates(sample_dict)
print("Dictionary after removing duplicates:", result)
```

Output 10:

```
iversity\011 Python\new.py  
Difference in list1: [1, 2]  
Difference in list2: [5, 6]
```

Output 11: a)

```
Dictionary after removing duplicates: {'Manjeet': [1  
, 'Akash': []}
```

11. b) Write a Python program to count the frequencies in a list using a dictionary.

```
def count_frequencies(lst):  
    frequency_dict = {}  
    for item in lst:  
        # Check if the item is already in the dictionary  
        if item in frequency_dict:  
            frequency_dict[item] += 1  
        else:  
            frequency_dict[item] = 1  
    return frequency_dict
```

```
# Example usage  
input_list = [1, 1, 1, 5, 5, 3, 1, 3, 3, 1, 4, 4, 4, 2, 2, 2, 2]  
result = count_frequencies(input_list)  
print("Frequencies:", result)
```

Problem 12. a) Write a Python Program to Capitalize the First Letter of Each Word in a File.

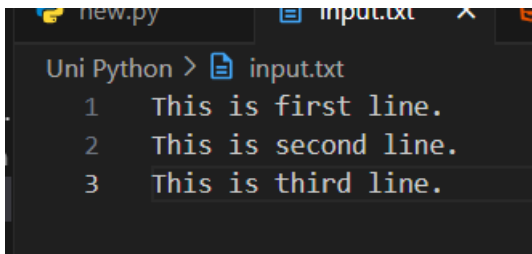
```
# Program to capitalize the first letter of each word in a file  
def capitalize_file_words(file_path):  
    with open(file_path, 'r') as file:  
        content = file.read()  
        capitalized_content = ' '.join(word.capitalize() for word in content.split())  
    return capitalized_content
```

```
# Example usage  
# Assuming 'input.txt' contains the text  
file_path = 'input.txt'  
result = capitalize_file_words(file_path)  
print("Capitalized content:", result)
```

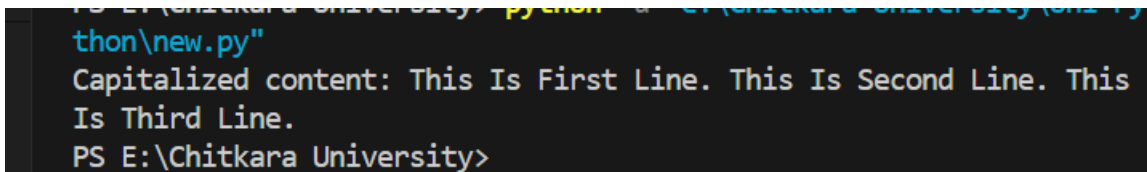
Output 11: b)

```
Frequencies: {1: 5, 5: 2, 3: 3, 4: 3, 2: 4}
```

Output 12(depends on content of 'input.txt'): a)



```
Uni Python > input.txt
1 This is first line.
2 This is second line.
3 This is third line.
```



```
PS E:\Chitkara University> python "C:\Chitkara University\Chitkara University\Python\new.py"
Capitalized content: This Is First Line. This Is Second Line. This Is Third Line.
PS E:\Chitkara University>
```


Problem 12. b) Write a Python Program to Print the Contents of a File in Reverse Order.

```
# Program to print the contents of a file in reverse order
def print_file_reverse(file_path):
    with open(file_path, 'r') as file:
        content = file.readlines()
        for line in reversed(content):
            print(line.strip())

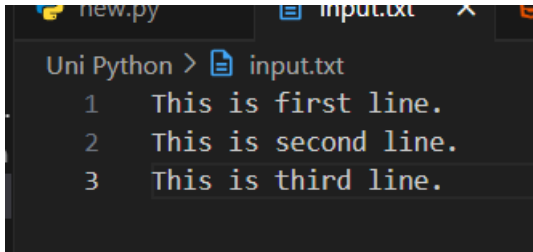
# Example usage
# Assuming 'input.txt' contains multiple lines
file_path = 'input.txt'
print_file_reverse(file_path)
```

Problem 13. Write a program to catch an exception and handle it using try and except code blocks.

```
# Program to demonstrate exception handling
def divide_numbers(a, b):
    try:
        result = a / b
    except ZeroDivisionError:
        return "Cannot divide by zero."
    return result

# Example usage
num1 = 10
num2 = 0
result = divide_numbers(num1, num2)
print("Result:", result)
```

Output 12(depends on content of 'input.txt'): b)

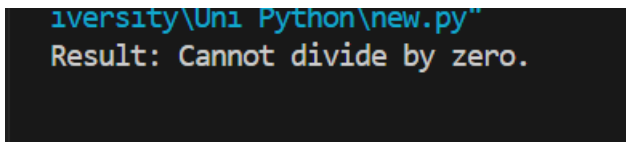


```
new.py  input.txt x
Uni Python > input.txt
1 This is first line.
2 This is second line.
3 This is third line.
```



```
> python -u "e:\Chitkara University\Uni Py
thon\new.py"
This is third line.
This is second line.
This is first line.
PS E:\Chitkara University>
```

Output 13:



```
iversity\Uni Python\new.py"
Result: Cannot divide by zero.
```

14. Write a Python Program to Append, Delete and Display Elements of a List using Classes.

Program to append, delete and display elements of a list using classes

```
class ListManager:
```

```
    def __init__(self):
```

```
        self.elements = []
```

```
    def append_element(self, element):
```

```
        self.elements.append(element)
```

```
    def delete_element(self, element):
```

```
        if element in self.elements:
```

```
            self.elements.remove(element)
```

```
        else:
```

```
            return "Element not found."
```

```
    def display_elements(self):
```

```
        return self.elements
```

Example usage

```
manager = ListManager()
```

```
manager.append_element(1)
```

```
manager.append_element(2)
```

```
manager.delete_element(1)
```

```
print("Current list:", manager.display_elements())
```

Output 14:

```
Current list: [2]
```

Problem 15: Write a Python Program to Find the Area and Perimeter of the Circle using Class.

```
# Program to find the area and perimeter of a circle using a class
import math
```

```
class Circle:
    def __init__(self, radius):
        self.radius = radius

    def area(self):
        return math.pi * (self.radius ** 2)

    def perimeter(self):
        return 2 * math.pi * self.radius
```

```
# Example usage
circle = Circle(5)
print("Area of the circle:", circle.area())
print("Perimeter of the circle:", circle.perimeter())
```

Output 15:

```
Area of the circle: 78.53981633974483  
Perimeter of the circle: 31.41592653589793  
PS E:\Chitkara University>
```

Problem 16: Create an interactive application using Python's Tkinter library for graphics programming.

```
import tkinter as tk
```

```
class CounterApp:
    def __init__(self, root):
        self.root = root
        self.root.title("Simple Counter App")

        # Initialize the counter value
        self.counter = 0

        # Label to display the counter
        self.counter_label = tk.Label(root, text="0", font=("Arial", 48))
        self.counter_label.pack(pady=20)

        # Increase button
        self.increase_button = tk.Button(root, text="Increase", command=self.increase, font=("Arial", 14))
        self.increase_button.pack(side=tk.LEFT, padx=20)

        # Decrease button
        self.decrease_button = tk.Button(root, text="Decrease", command=self.decrease, font=("Arial", 14))
        self.decrease_button.pack(side=tk.LEFT, padx=20)

        # Reset button
        self.reset_button = tk.Button(root, text="Reset", command=self.reset, font=("Arial", 14))
        self.reset_button.pack(side=tk.LEFT, padx=20)

    def increase(self):
        self.counter += 1
        self.update_counter()

    def decrease(self):
        self.counter -= 1
        self.update_counter()

    def reset(self):
        self.counter = 0
        self.update_counter()

    def update_counter(self):
        self.counter_label.config(text=str(self.counter))

# Create the Tkinter window and run the application
root = tk.Tk()
app = CounterApp(root)
root.mainloop()
```



Output 16:

